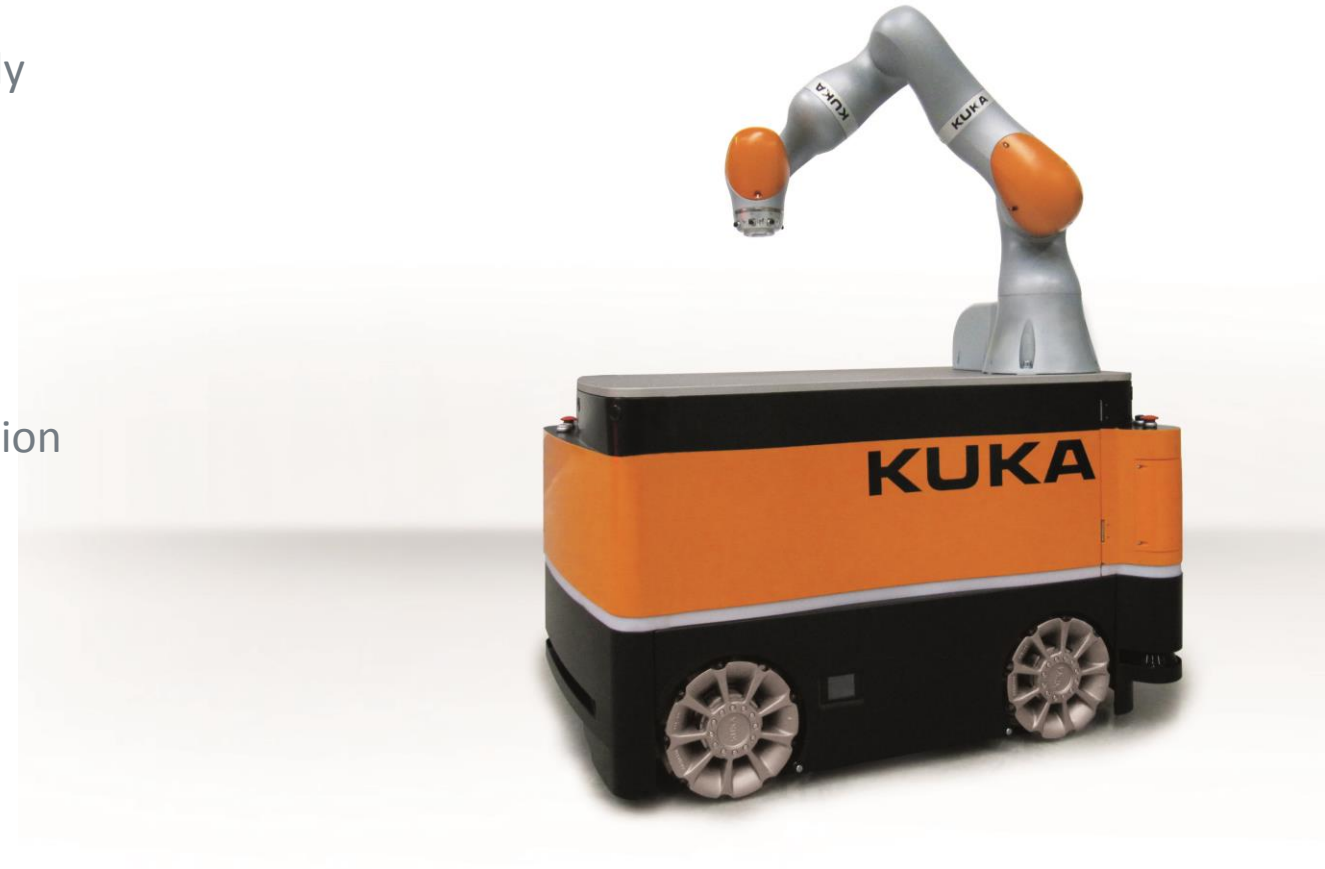
KUKA

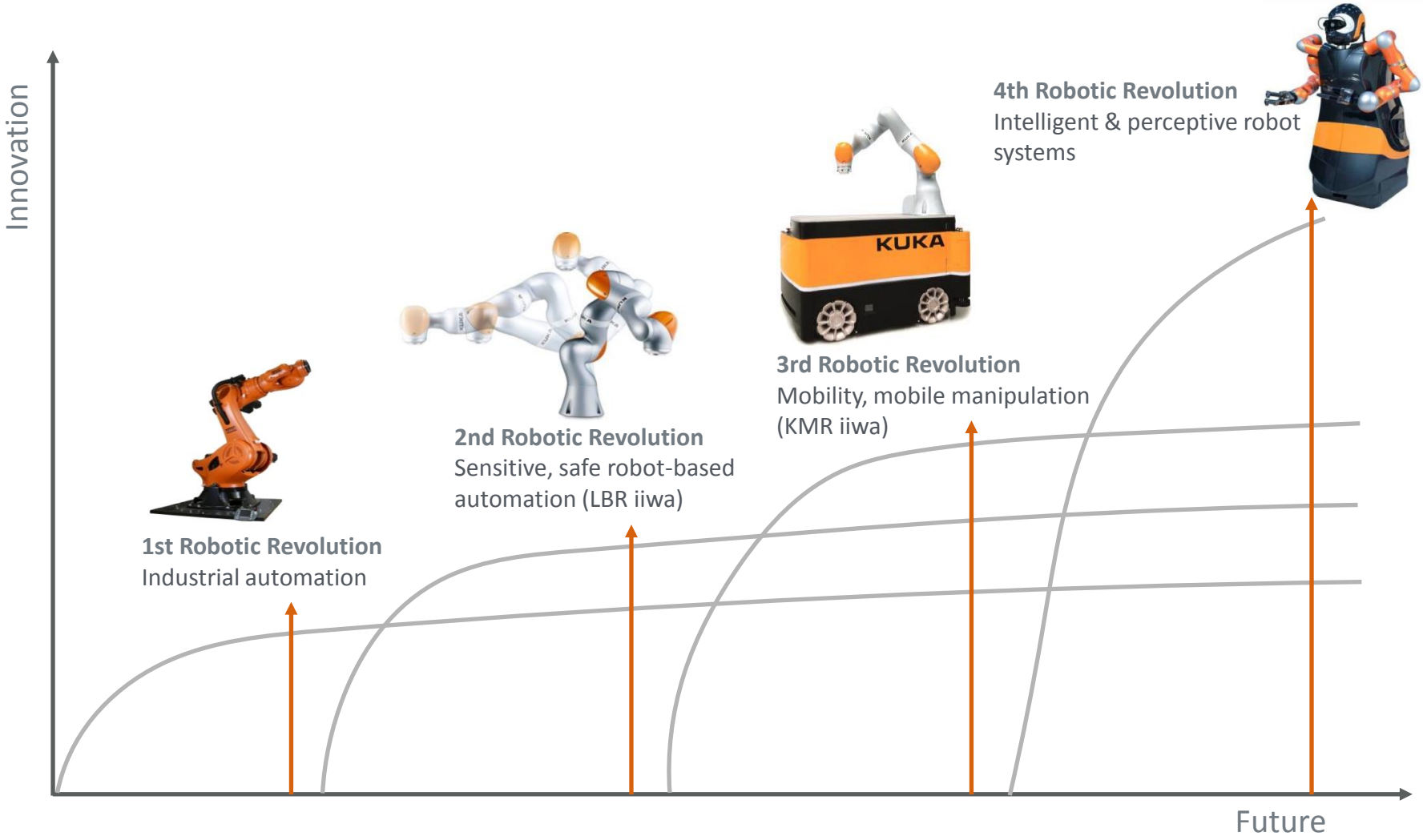
KMR iiwa – a new generation of mobile robotics

Content

1. Product presentation
   - The new product family
   - Functions
   - KMP omniMove
   - LBR iiwa – Robot arm
   - KUKA Sunrise
   - KUKA Navigation Solution
2. KMR iiwa advantages
3. Industry 4.0



KUKA products



KUKA Mobile Robotics

KMP - KUKA Mobile Platforms

Payload

< 400kg

KMP 400
(field test)

1,5t

KMP 1500
(in development)

> 3t

KMP omniMove
(available, project based)

KMR - KUKA Mobile Robots

Robot type

LBR iiwa

KMR iiwa
(field test)

Quantec

KMR Quantec
(available, project based)

All mobile robotics components out of one hand!

Robot

+

Mobile platform

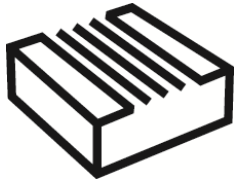
+

Controller

+

Software

+



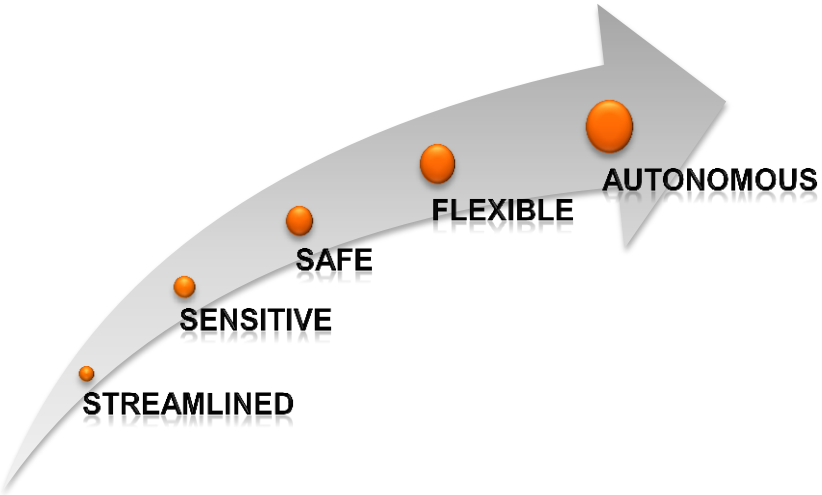
Navigation

Service-proven KUKA components for innovative production scenarios

KMR iiwa

What is KMR iiwa?

- KMR iiwa stands for **K**UKA **M**obile **R**obotics **iiwa**
- Combination of two existing KUKA products: LBR iiwa and omniMove platform
- HRC – Human Robot Collaboration
- Intelligent mobile assistant for future versatile production concepts



The new product family

New mobile Platform
KMP omniMove 400

All in one control based on extended
Sunrise controller

LBR iiwa robot arm in two payload
variants: 7 and 14 kg

Autonomous navigation
KUKA Navigation Solution

Controller operating system
Sunrise.OS

Sunrise.Workbench for configuration
and programming

KMR iiwa functions

Autonomous navigation: KMR iiwa moves autonomously without risk of collision

- Safe laser scanner for environment monitoring
- KUKA Navigation Solution offers a trajectory path planner for collision-free path planning

360°

Maximum flexibility and unrestricted maneuverability

- Omnidirectional mobile platform for limitless adaptability
- Null space motion for the robot arm to reach any position in workspace
- Cost-effectiveness in logistics management

± 1 mm

Utmost precision and simple operator control

- Fine localization feature in the navigation software
- Omnidirectional wheel technology enables “KMR iiwa” to maneuver safely to the desired position in confined spaces

KMR iiwa functions

HRC able - Safe work between human and robots

- Integrated joint torque sensors in safe technology
- Safety-oriented position sensors
- Freely configurable safety controller for application risk assessment

Freely scalable modular system

- Position and number of installed robots is variable
- Extend the mobile platform with application layer for external computer, all necessary hardware to fulfill application requirements

KMR iiwa control layer

[+] Wireless Joystick with integrated safe emergency button

[+] All driven by one controller
Extended Sunrise Controller

[+] Wireless communication
Configuration & programming

[+] Integrated navigation PC
KUKA Nav-Box

KMP omniMove safety features

[+] Led-sound buzzer

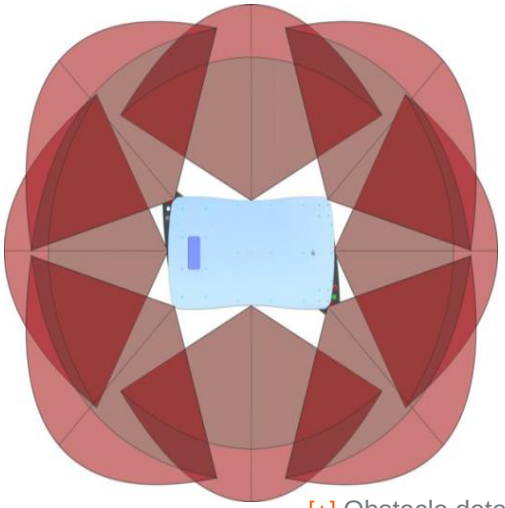
[+] 2 x Emergency Stop

8 x Ultrasound sensor [+]

[+] RGB Led band

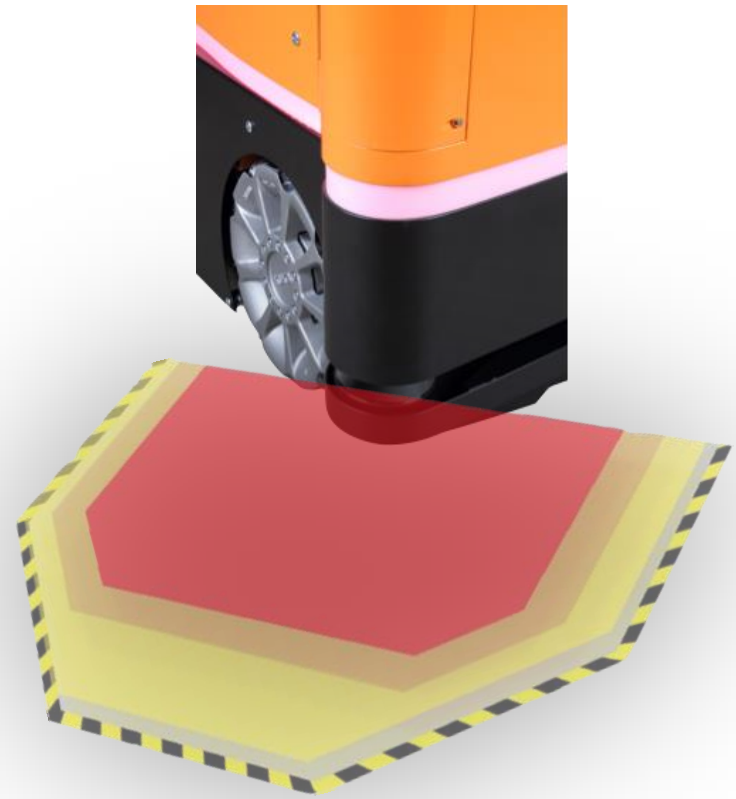
2 x Laser scanner [+]

KMP omniMove safety features



[+] Obstacle detection using ultrasound sensors

[+] Protective safety fields
Warning fields



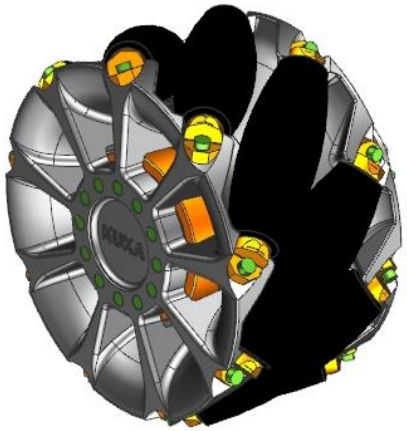
[+] Visual signal

Existing product – The Mecanum wheel

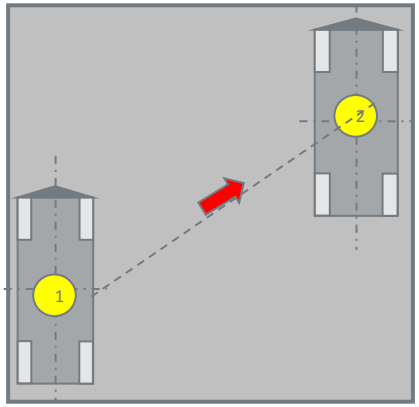
On all our existing platforms we use our Mecanum-wheel!

- The **omniMove®** (Mecanum) wheel has two fixed rims and spherical rolls
- The rolls are fixed in a 45° angle to the rims
- The whole wheel is electric powered
- All rolls are free rotating

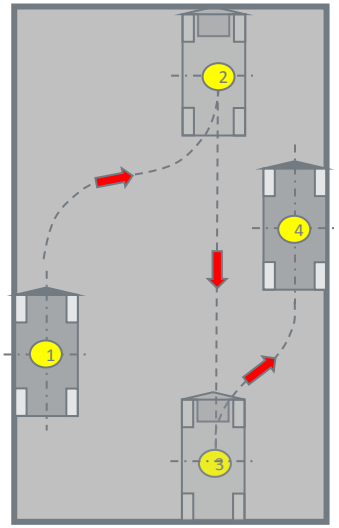
- unlimited maneuverability
- reduction of logistic area (= more production area)
- no special floor preparation (standard concrete)
- no wear of the floor
- very high positioning accuracy



With **omniMove®** wheels

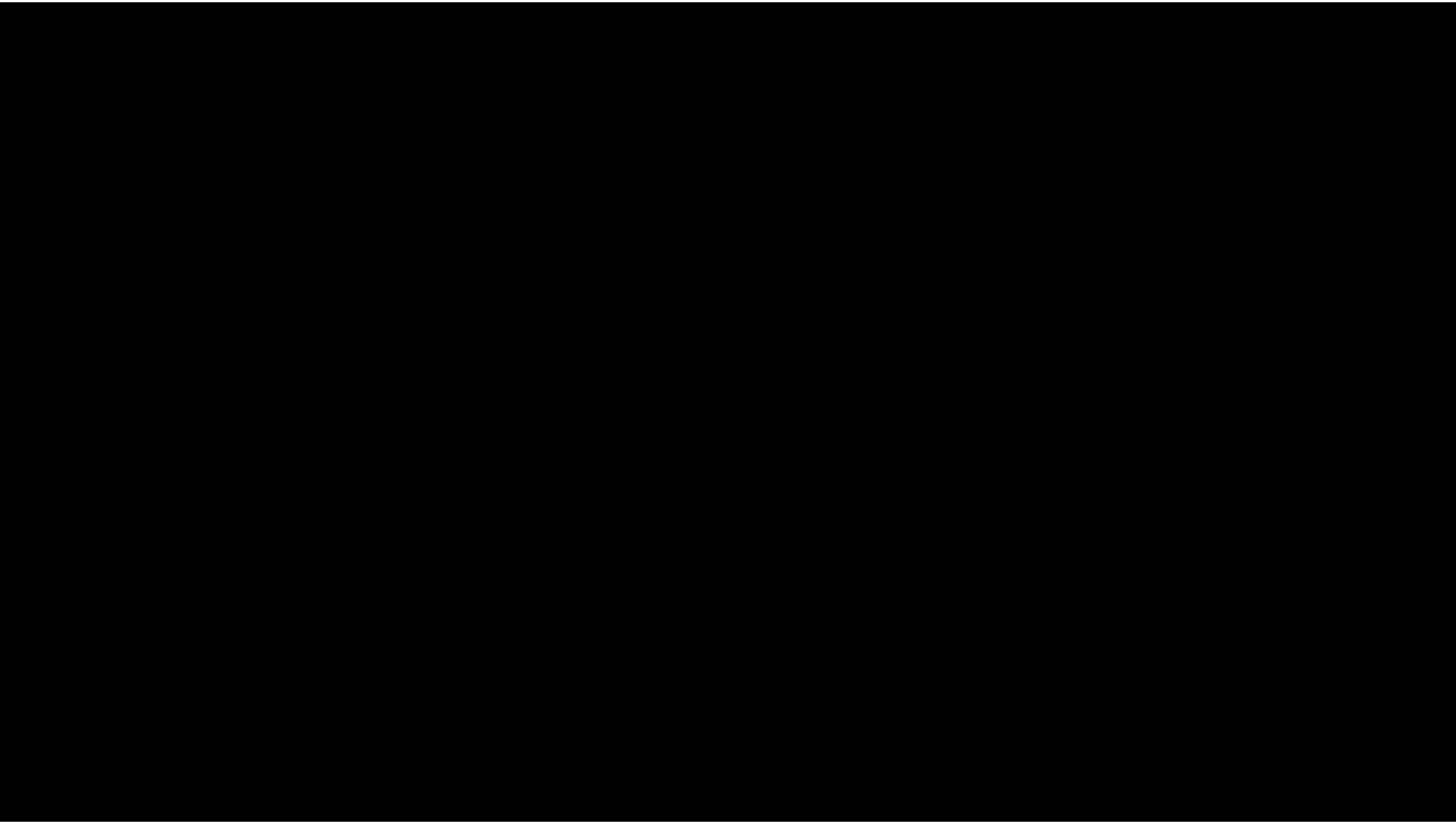


➡ Reduction of maneuvering area in this case approx. 50%

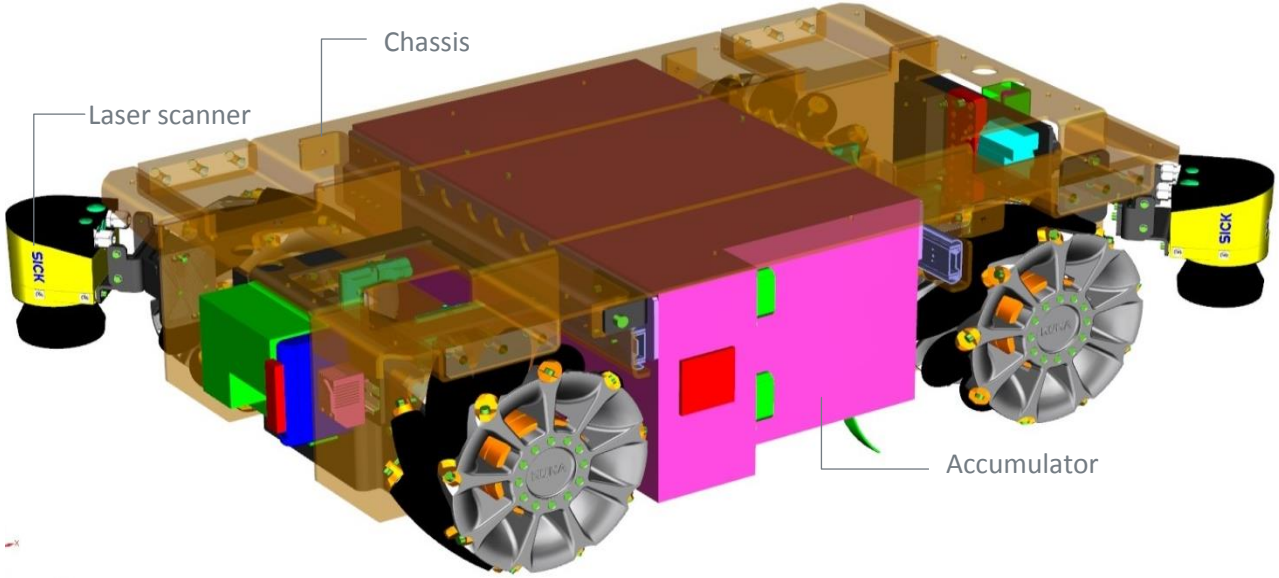


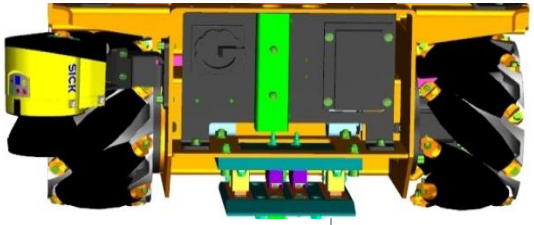
With conventionally steered wheels

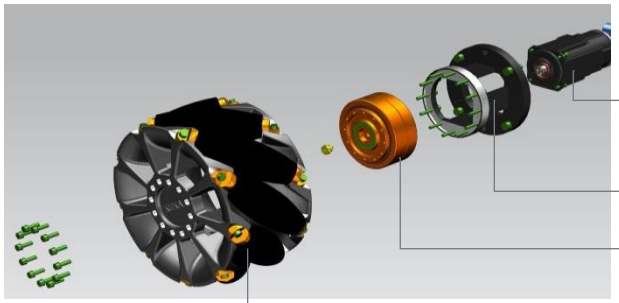
Existing product – The Mecanum wheel



KMP omniMove power & drives



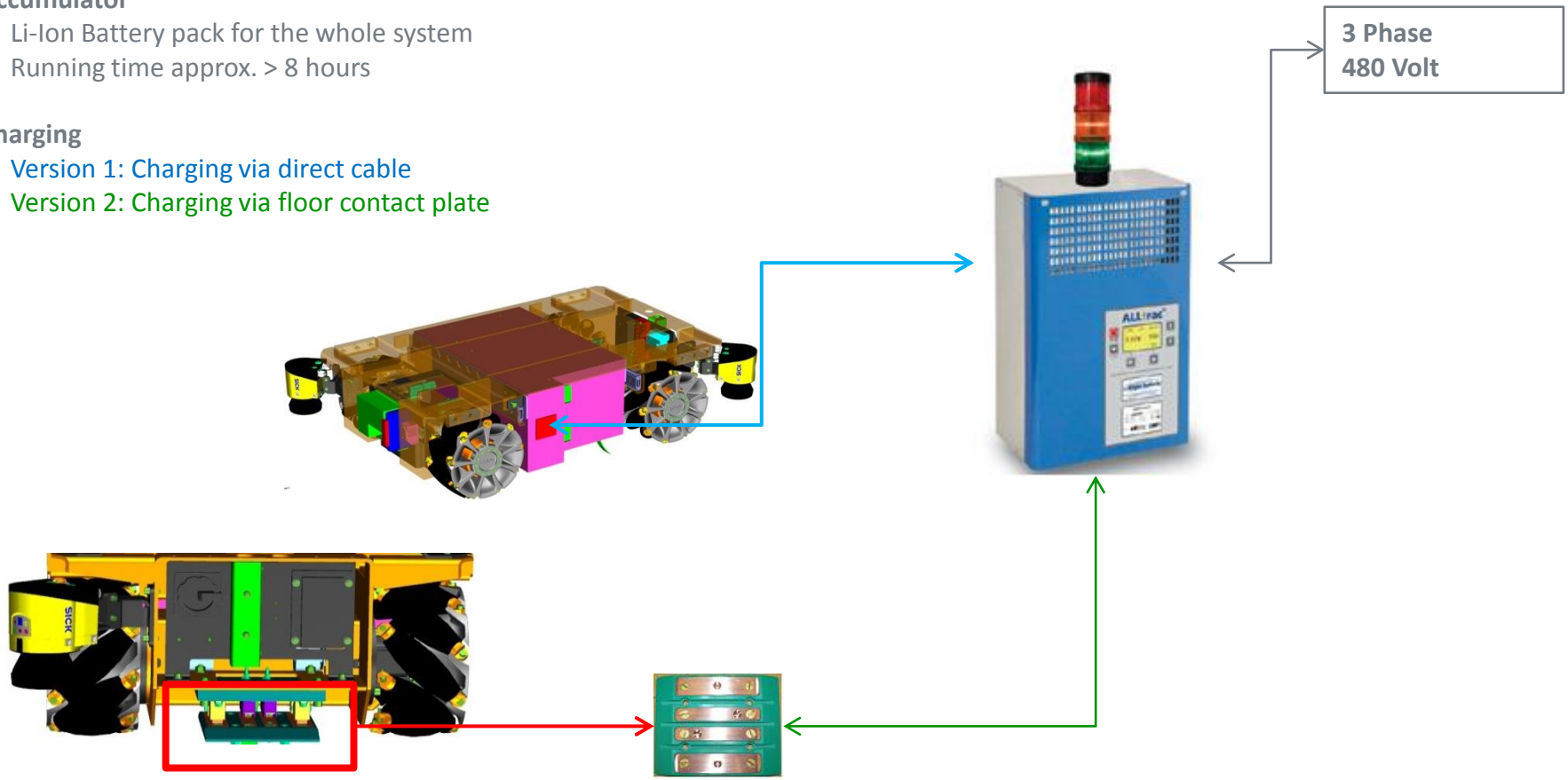




KMP 400 Charging concept

Accumulator
- Li-Ion Battery pack for the whole system
- Running time approx. > 8 hours

Charging
- Version 1: Charging via direct cable
- Version 2: Charging via floor contact plate



KMP omniMove 400 technical details

Features	Dimension	Unit
Height of the vehicle	700	mm
Length with scanners	1190	mm
Width with scanners	720	mm
Weight approx.	400	kg
Additional payload	Max. 200	kg
Maximum velocity straight direction	3	km/h
Maximum velocity crosswise direction	2	km/h
Wheel diameter	250	mm
Surrounding temperature	+5°C…+35°C	
Humidity	max. 80%	

Product overview – KUKA Extended Sunrise Cabinet

- Based on KUKA „Sunrise cabinet“ technology
- High-speed servo control
- Integrated and configurable software safety controller
- Compact design (19")
- All driven by one controller (robot arm + mobile platform)
- Supports all common field bus systems

	KUKA Extended Sunrise Cabinet
Processor	Quadcore-Processor
Hard drive	SSD
Interfaces	USB, Ethernet, DVI-I
Protection level	IP 20
Dimensions (LxWxH)	500 mm x 483 mm x 278 mm
Weight	33 kg

LBR iiwa

LBR iiwa is an acronym and stands for :

(i) **i**ntelligent
(i) **i**ndustrial
(w) **w**ork
(a) **a**ssistant

An intelligent assistant for future-oriented production concepts

1. … is sensitive
2. … is safe HRC – able
3. … has 7 axes
4. … is light

LBR iiwa – The features

New era in robotics …

Teaching by demonstration [+]

[+] Sensitive assembly processes

Bionic motions using 7 axes [+]

[+] Safety

[+] Sensitivity

Flexibility [+]

[+] Compliance mode

[+] Integrated torque sensors

Product overview – LBR iiwa, 7 kg and 14 kg

For 2015 pre-serial KMR iiwa versions:
Only LBR iiwa 14 R800 with MF electrical available

LBR iiwa	7 R800	14 R820
Rated payload	7 kg	14 kg
Weight	23,9 kg	29,9kg
Number of axes	7	7
Protection level	IP 54	IP 54
Pose repeatability	±0,1 mm	±0,1 mm

Media-Flange variations

Basic-Flange		
	Electronic	Pneumatic
Media-Flange		
Media-Flange IO		
Media-Flange Touch		

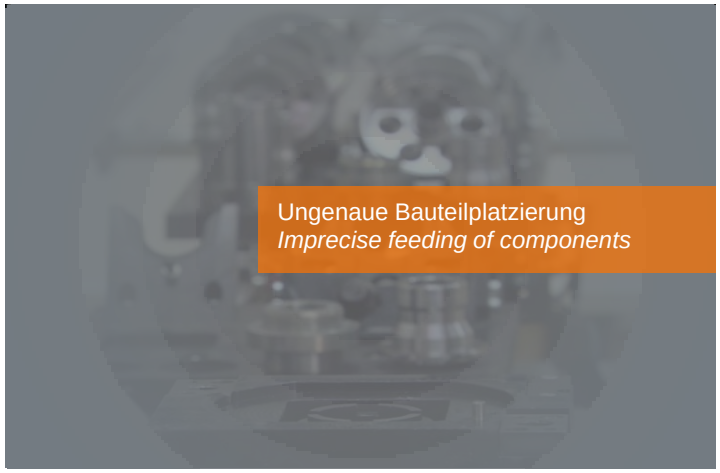
The new modes – Compliance mode / Gravity compensation

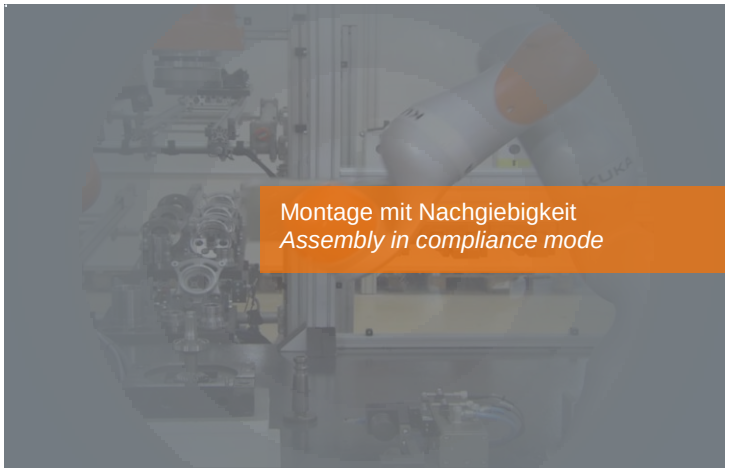
- The human acts, the LBR iiwa reacts
- The robot‘s behaviour can be easily programmed via the software
- Flexible adaption of the robot’s characteristics to the individual task
- Tasks are solved by the means of compliance rather than programmed positioning

- The LBR iiwa is weightless
- External gravitation is compensated via the integrated sensors and the controller
- Only small application of force is necessary for moving the robot in a new pose
- Is basis for features like hand guiding and teaching by demonstration

The features – Sensitive assembly processes

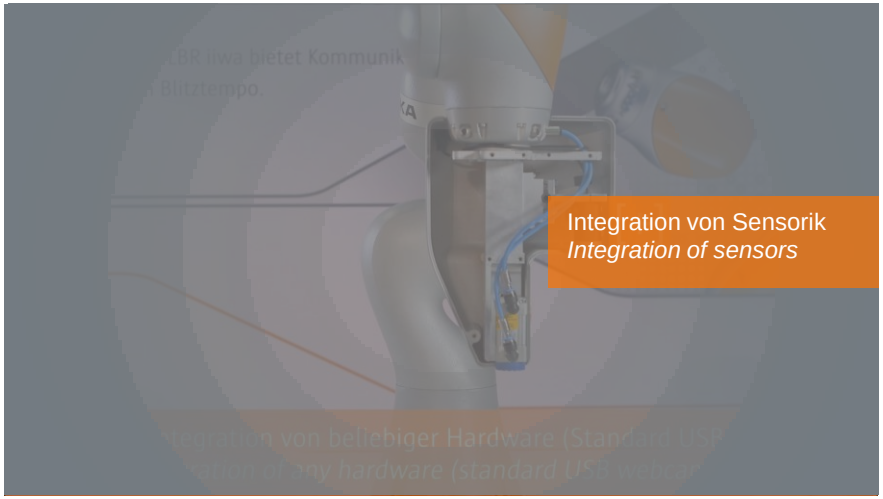
- LBR iiwa offers and integrates the sensitivity for new assembly processes
- Simple programming for joining processes
- LBR iiwa integrates the compliance in the structure instead
- Reduction of costs and cycle time in engineering, manufacturing and commissioning
- Applications with integrated tactile search without vision systems
- Allows imprecise feeding of components





Product overview – KUKA Sunrise.OS

- The KUKA Sunrise.OS (Operating System) is the new operating system.
- KUKA Sunrise.OS – the technological platform for innovative robotics:
  - Multi-kinematics
  - Human-robot collaboration
  - Integration of sensors
  - Object-oriented programming



Programming robots with Java-technology

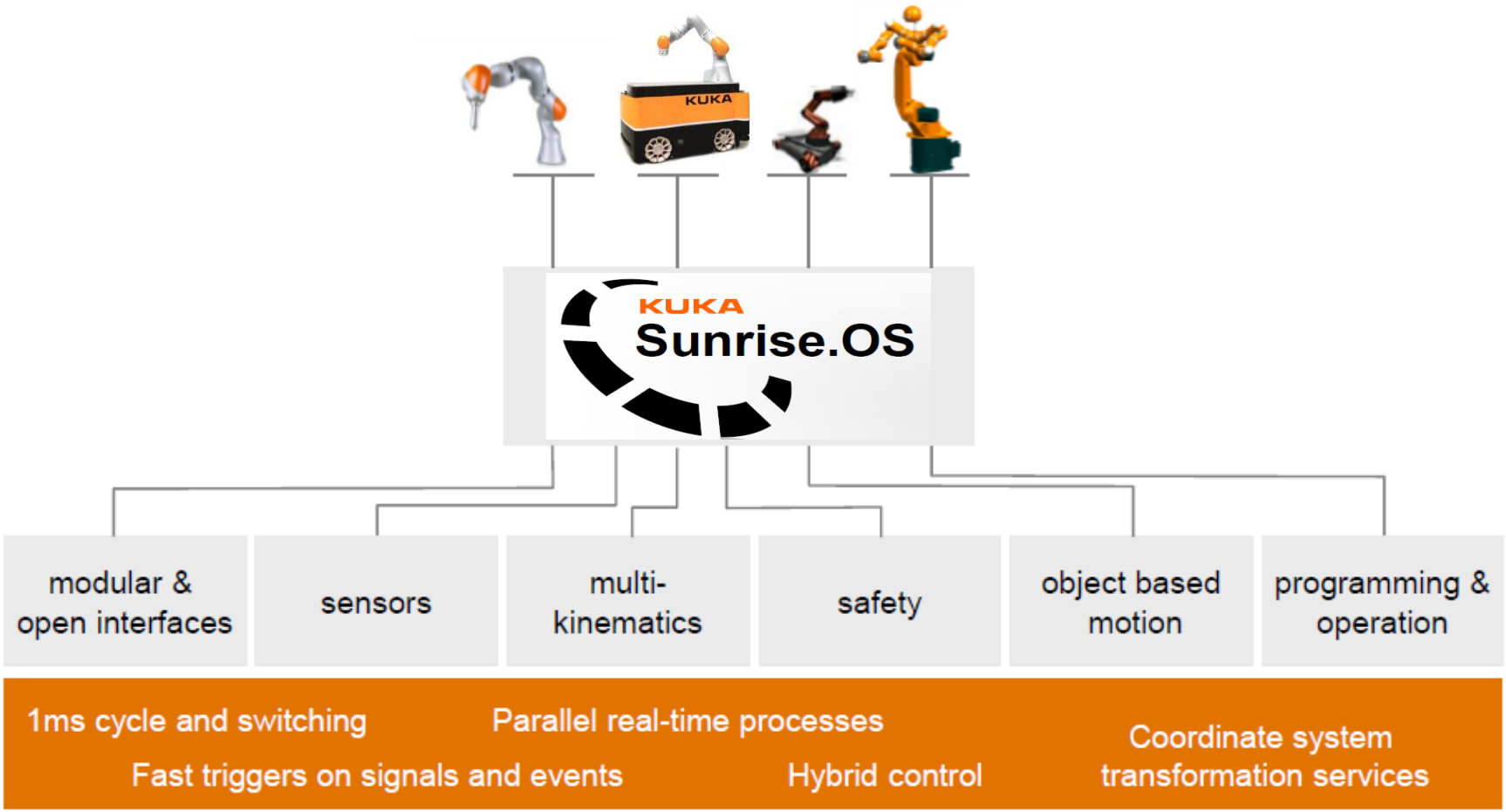
- State of the art programming language
- Platform independent
- Multithreading
- Object-oriented programming

3 Billion Devices Run Java

Computers, Printers, Routers, Cell Phones, BlackBerry, Kindle, Parking Meters, Public Transportation Passes, ATMs, Credit Cards, Home Security Systems, Cable Boxes, TVs...

ORACLE

Sunrise OS enables safe & sensor-based multi-kinematics

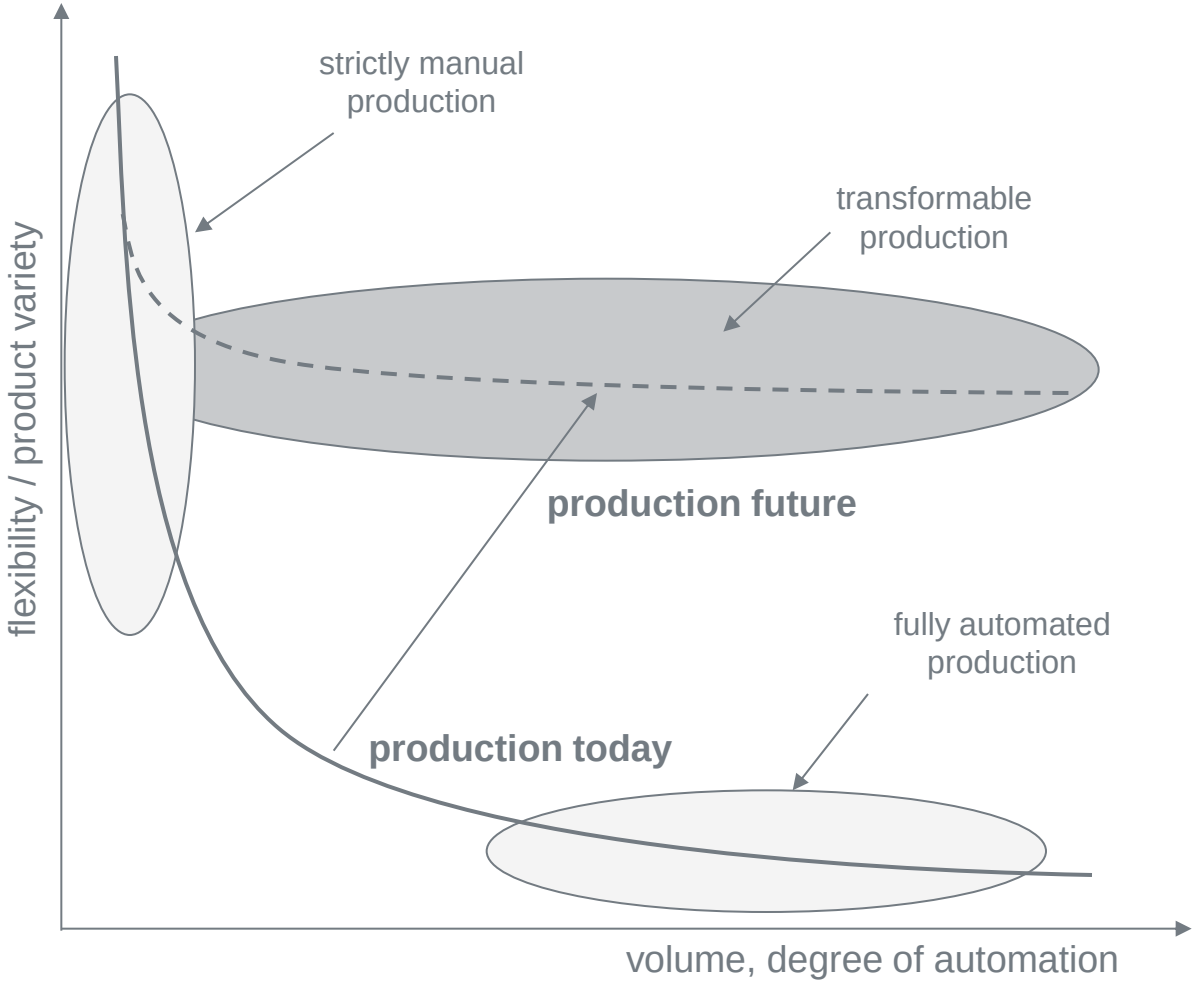


KUKA Sunrise.Workbench

- Eclipse-based engineering suite
- One tool for administration and programming
- State of the Art Editor with integrated documentation
- Expandability (Plug-In) by KUKA, customer or partner
- Easy-to-use user interface
- Object-oriented programming
- Simple project management and customization
- Fast start-up procedure
- Powerful diagnostics

KUKA Sunrise.OS

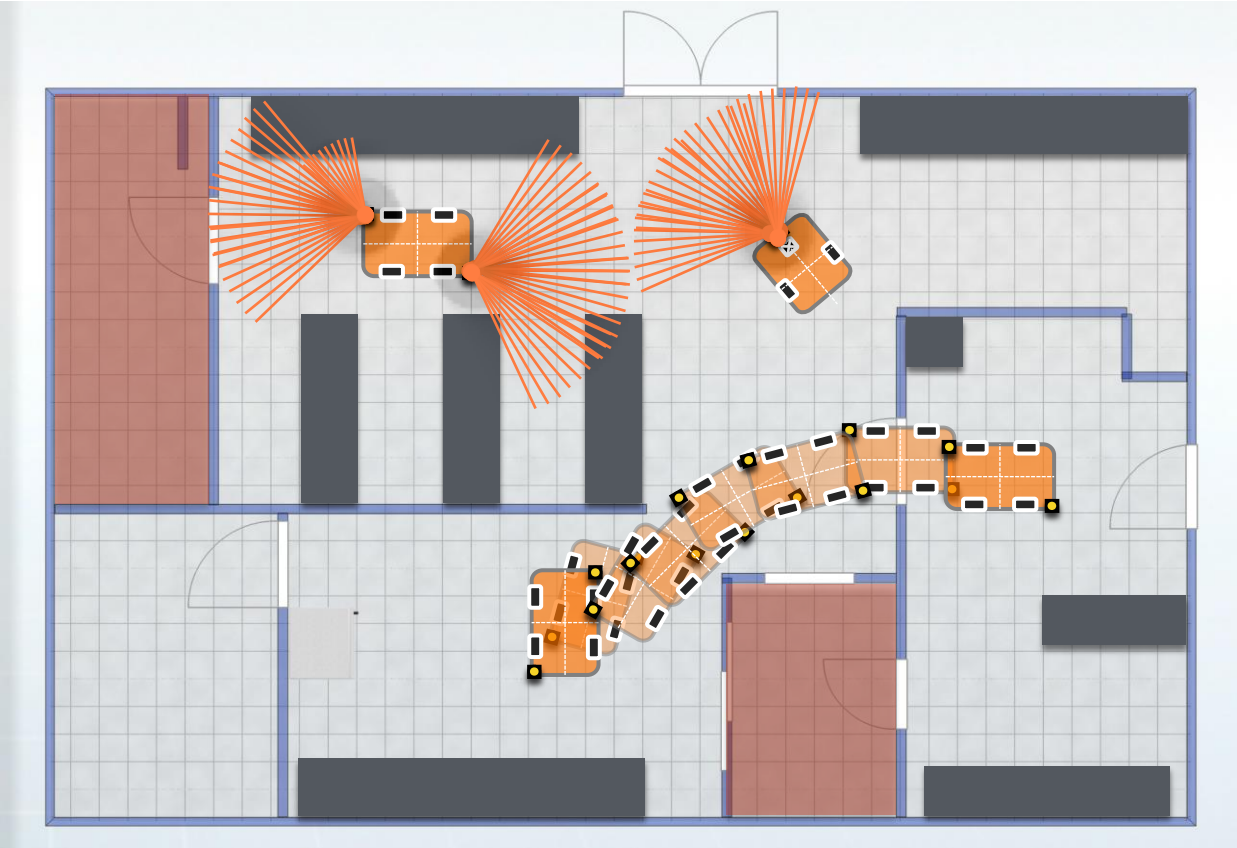
Motivation for autonomous navigation



modern days requirements in production areas:

- High flexibility and product variety at low and high volumes
- Low effort to change production
- Flexible degrees of automation

KUKA Autonomous Navigation Solution included



[+] Safety

[+] Flexibility

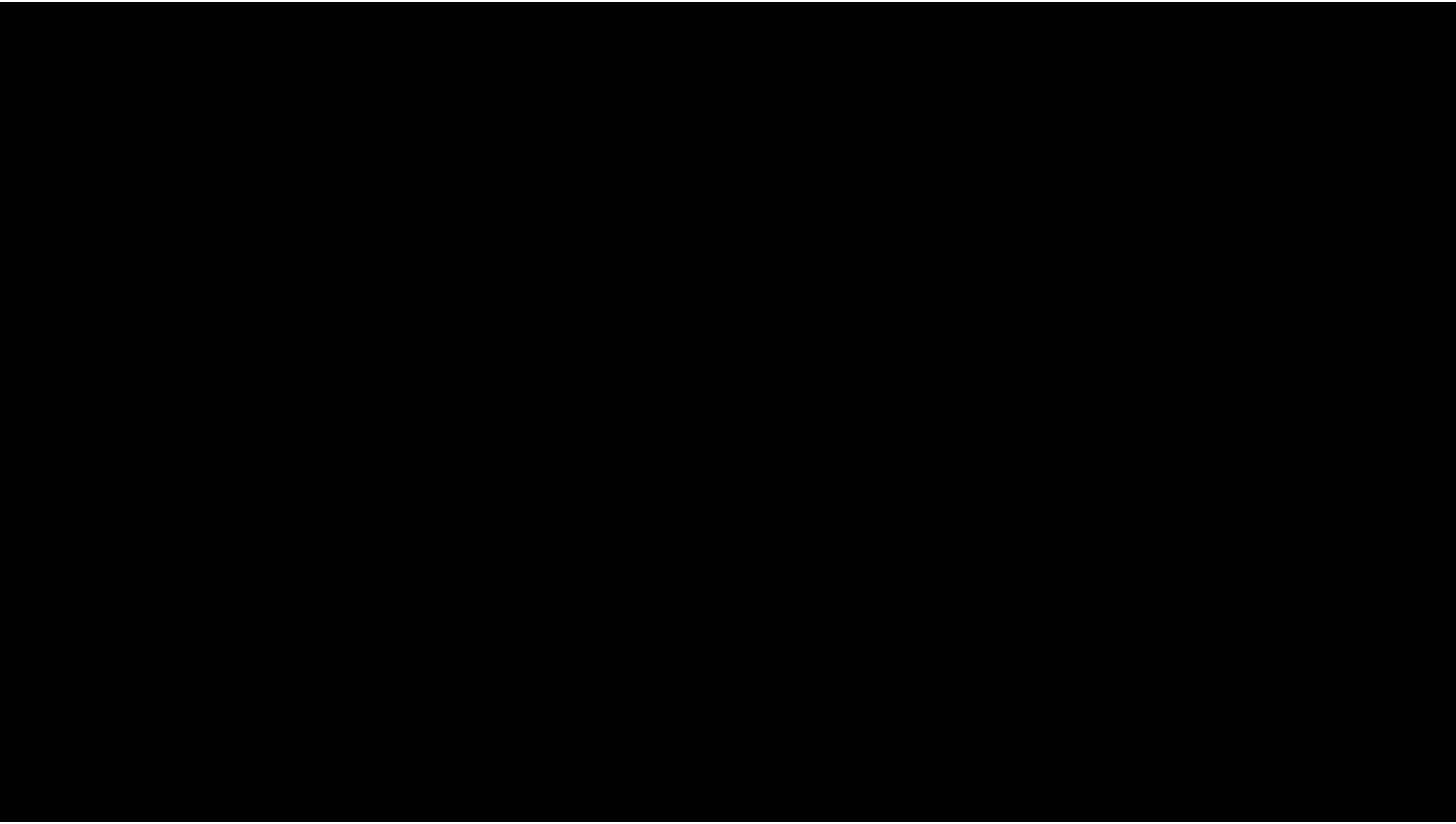
[+] Fleet Management

[+] Real-time control

[+] Collision-free path planning

[+] Scalable Autonomy

KUKA.NavigationSolution



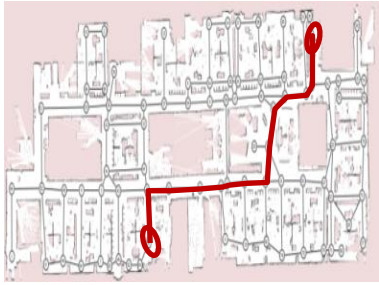
Basic questions for navigation

- How does the world look like? → Mapping (Graph-Based SLAM)

- Where am I? → Localization (Particle filter)

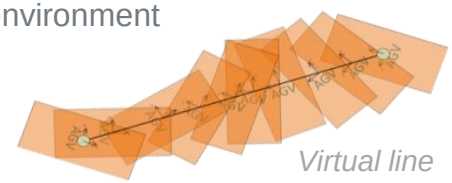
Simultaneous Localization and Mapping (SLAM)

- How to get from „A“ to „B“? → Path Planning

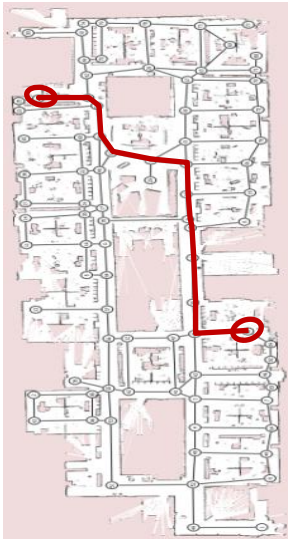


KUKA Autonomous Navigation Solution

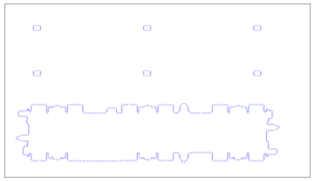
- Simultaneous Localization and Mapping (SLAM)
  - Based on laser and wheel encoder data
  - No need for artificial markers or changes to the environment
  - Robust against moving objects

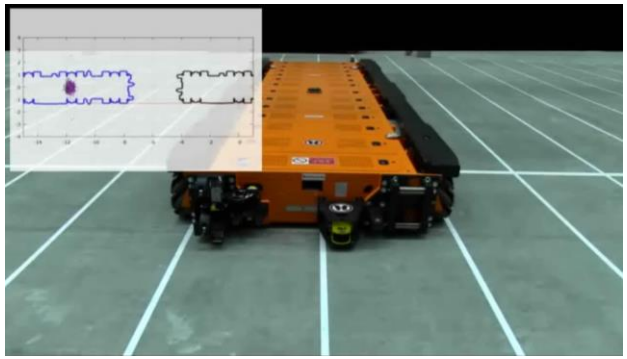


- Navigation with different degrees of autonomy
  - Fully-autonomous driving → Maximal flexibility
  - Semi-autonomous driving over virtual paths → Maximal predictability



- Object detection and tracking
  - Laser-based detection of any objects
  - Relative positioning to objects
  - High positioning accuracy





KUKA Autonomous Navigation Solution

Transformability

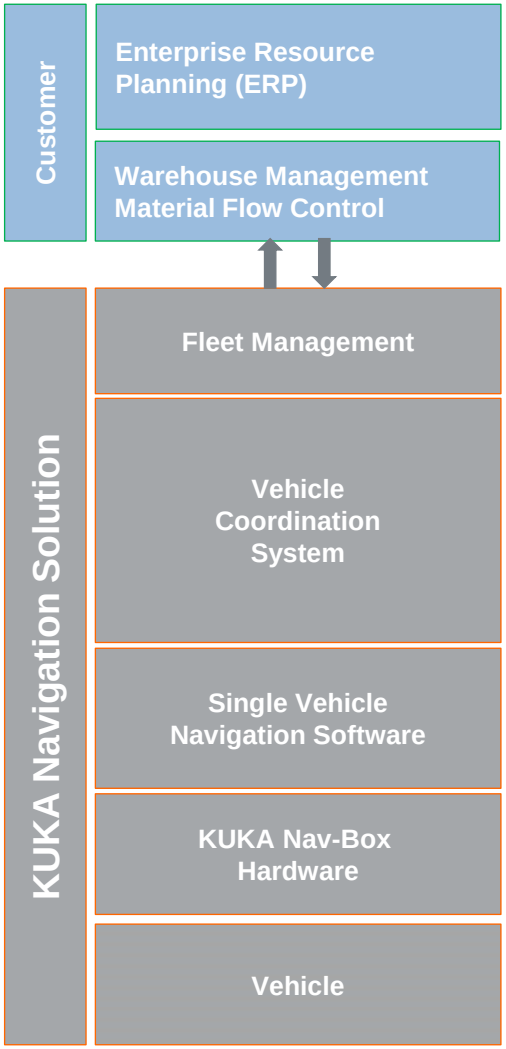
- fast, active adaptation of structures to unforeseen changes
- possibility to evolve structures responding to foreseeable and constantly changing requirements

Localization accuracy

- Long term test with dynamic environment for Aerospace Project
- average localization inaccuracy < 1,3 mm
- 95% of all measures inaccuracy < 3mm

SLAM is robust in cluttered environment!

KUKA Navigation Solution (Products)



Fleet Management

Traffic management & order planning

- The software is running on a central computer with interfaces to logistics and material flow system. Disposition, Planning and Scheduling
- High level planning and scheduling

Vehicle Coordination System (VCS)

- Software on a central computer for vehicle management, maintenance, data management and programming
- Interface to the individual vehicles

Vehicle Navigation Features

- **Mapping:** consistence, robustness, scalability and domain independence
- **Localization:** robustness and accuracy without environmental adaptations
- **Path Planning:** Scalable autonomy level depending on application and customer requirements
  Tracking: Relative positioning to known objects based on laser sensor data
- **3D sensors:** sensor data integration for obstacle detecting

KUKA Nav-Box HW

- Standardized industrial PC hardware with software interfaces for vehicle-independent use

KUKA Navigation Solution : Fleet Management

Customer

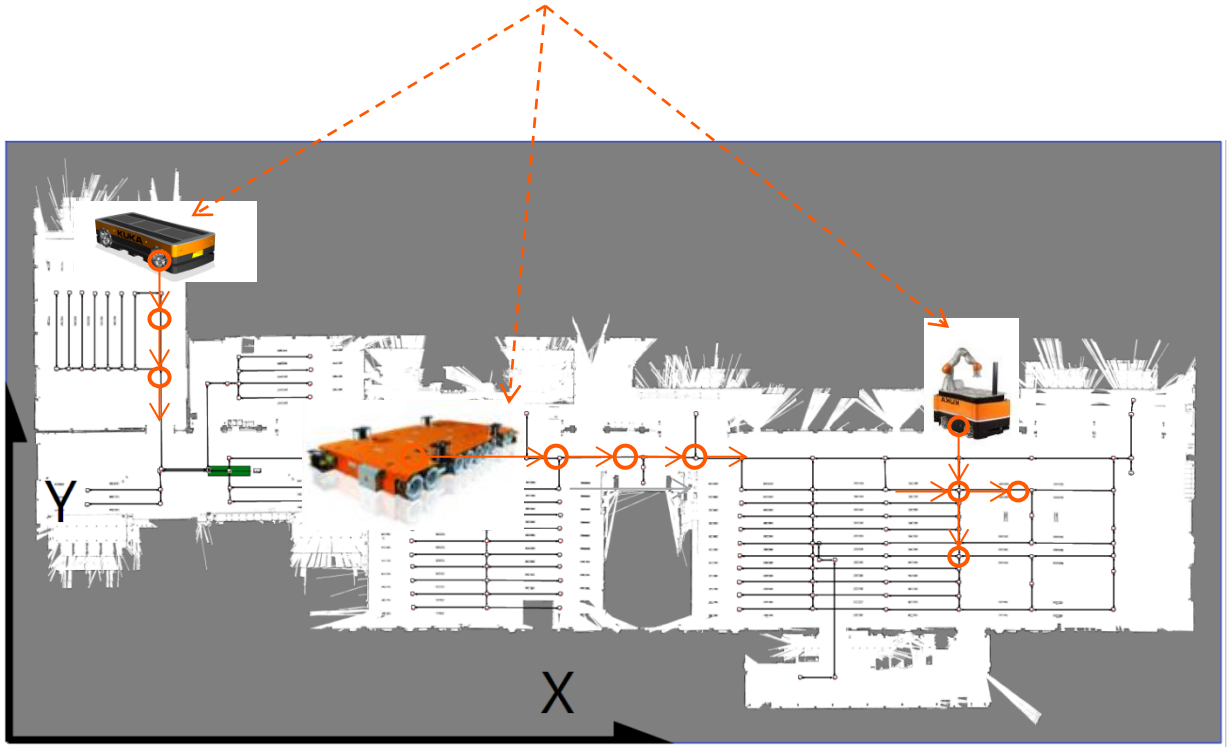
- Enterprise Resource Planning (ERP)
- Warehouse Management Material Flow Control

KUKA Navigation Solution

- Fleet Management
- Vehicle Coordination System
- Single Vehicle Navigation Software
- KUKA Nav-Box Hardware
- Vehicle

Traffic management & order planning

- The software is running on a central computer with interfaces to logistics and material flow system. Disposition, Planning and Scheduling
- High level planning and scheduling



Integration example

Enterprise Resource Planning
Warehouse Management System

TCP/IP : Port 5000

SECS Protocol

```
S2F41: 'S2F41' W
  <L [2]
    <A [5] 'ABORT'>
    <L [1]
      <L [2]
        <A [5] 'JOBID'>
        <A [4] 'Job1'>
      >
    >
  > .
```

SECS Parser
(From 3rd party SW Supplier)

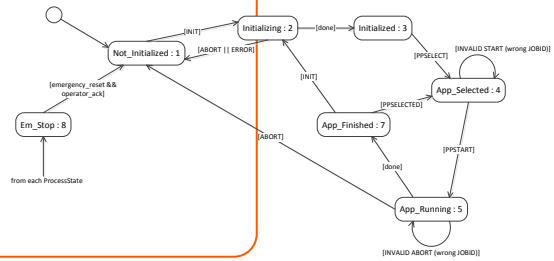
TCP/IP : Port 6000

Customized String

ABORT:Job1

Robotics API JAVA

Implement application taking in consideration different process/Job states : PPSELECT, INIT, START, ABORT, OnStateChange,…etc.



Optical data transmission

E23 / E84

Process equipment

Ready Plug & Play solution
(From 3rd party Supplier, for example:
Easy e23/e84 → Objective Solutions)

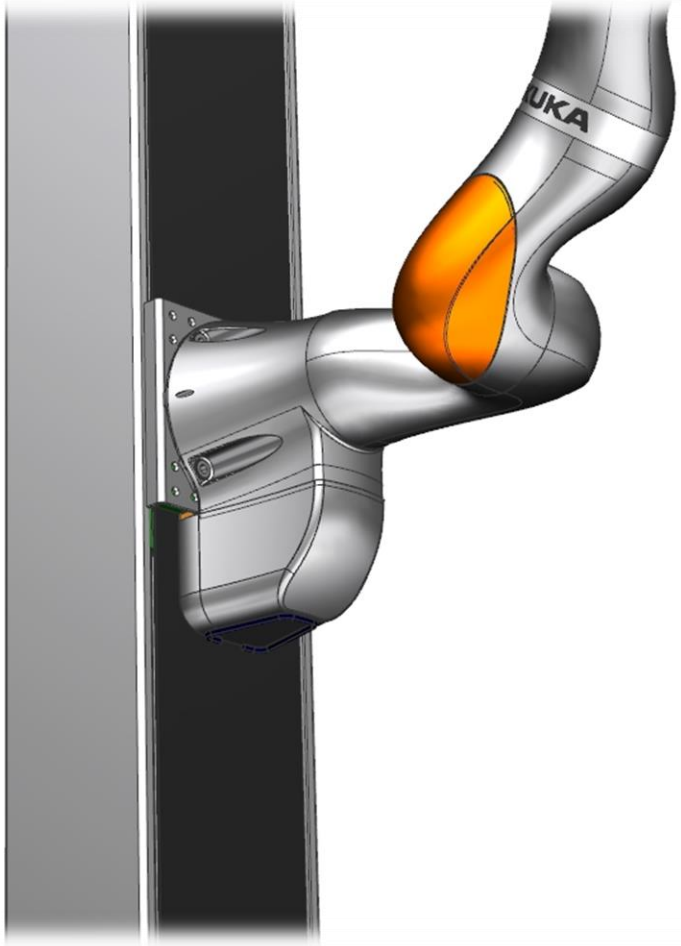
KMR iiwa pre-serial versions

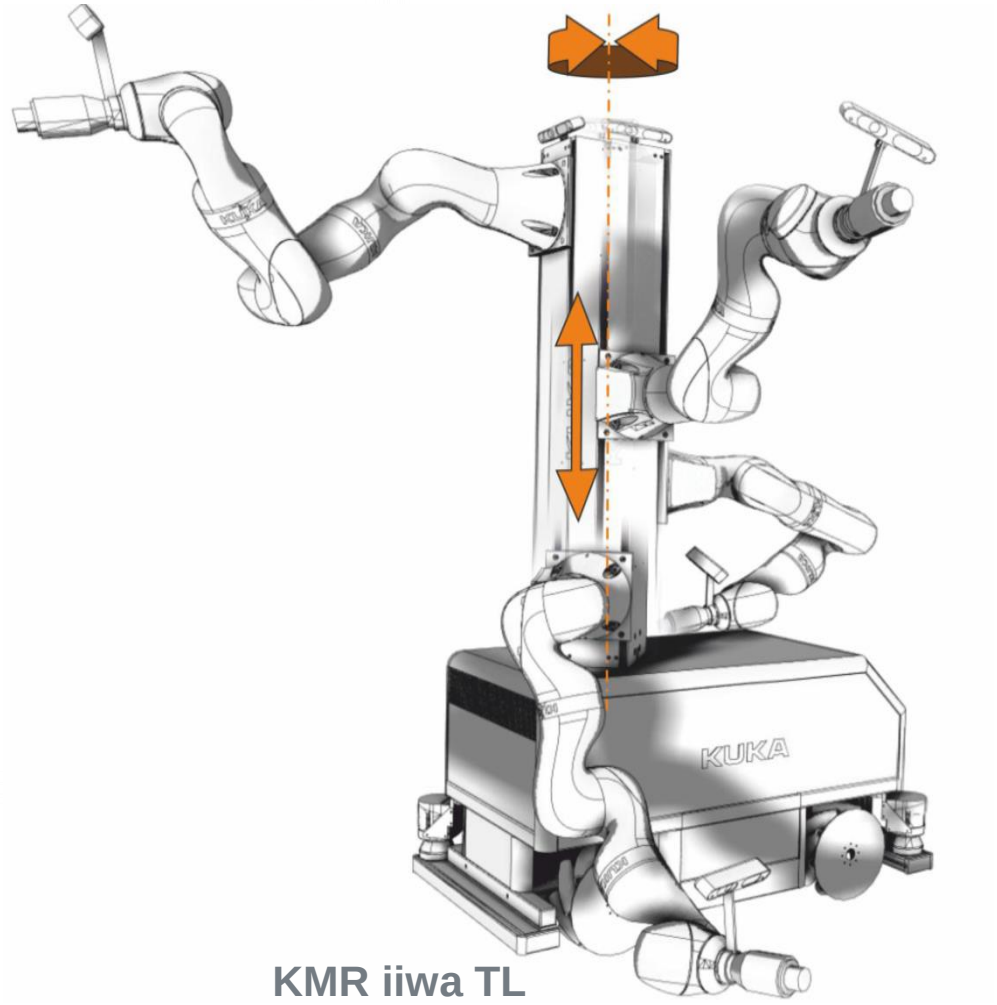
Application layer

Controller layer

Power and drives

KMR / KMP future development



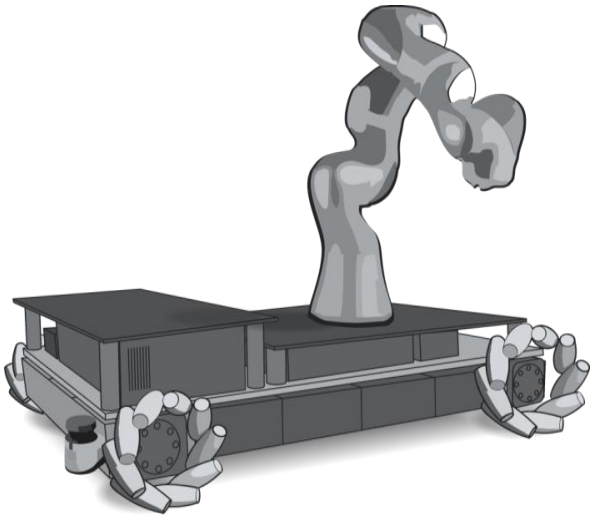


KMR iiwa TL

KMR / KMP future development

KMR iiwa Dual

KMR Agilus



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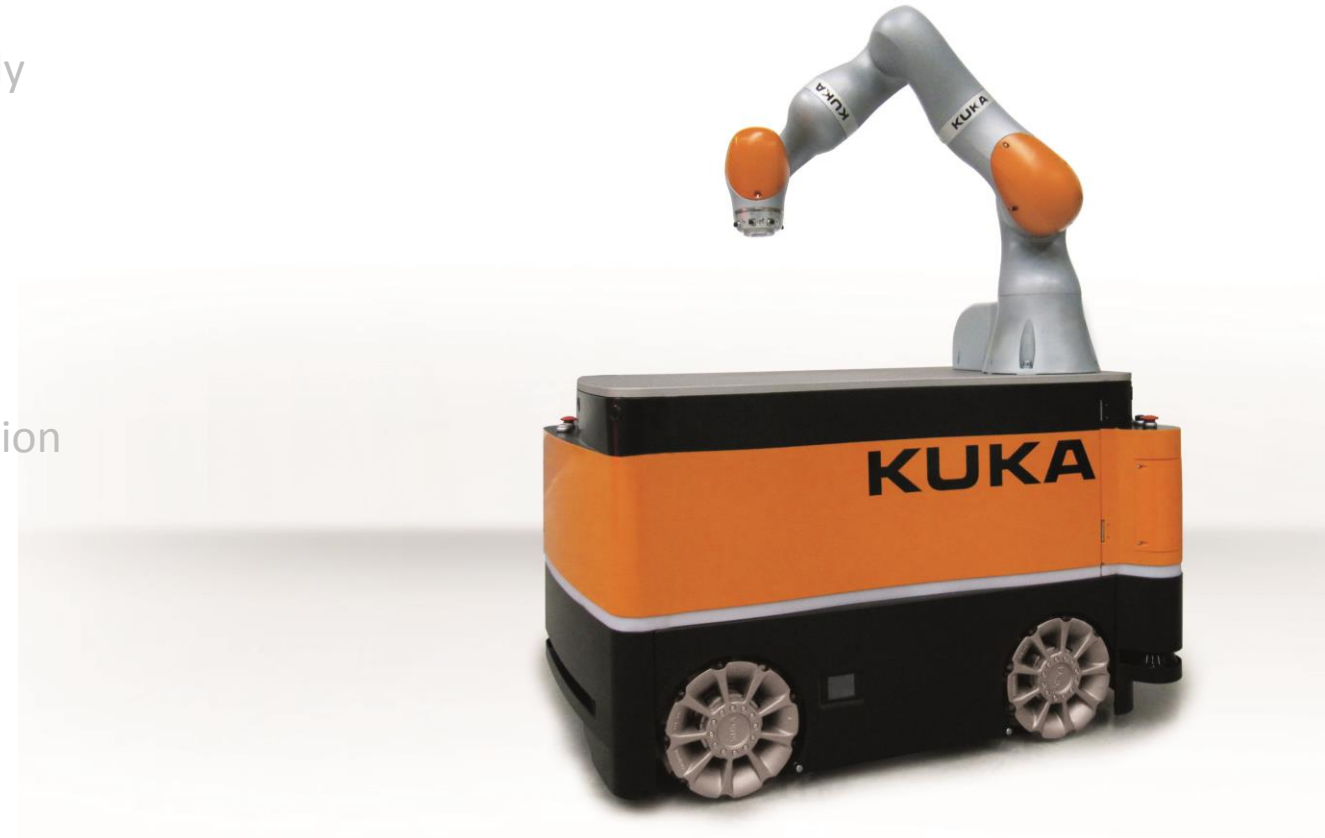
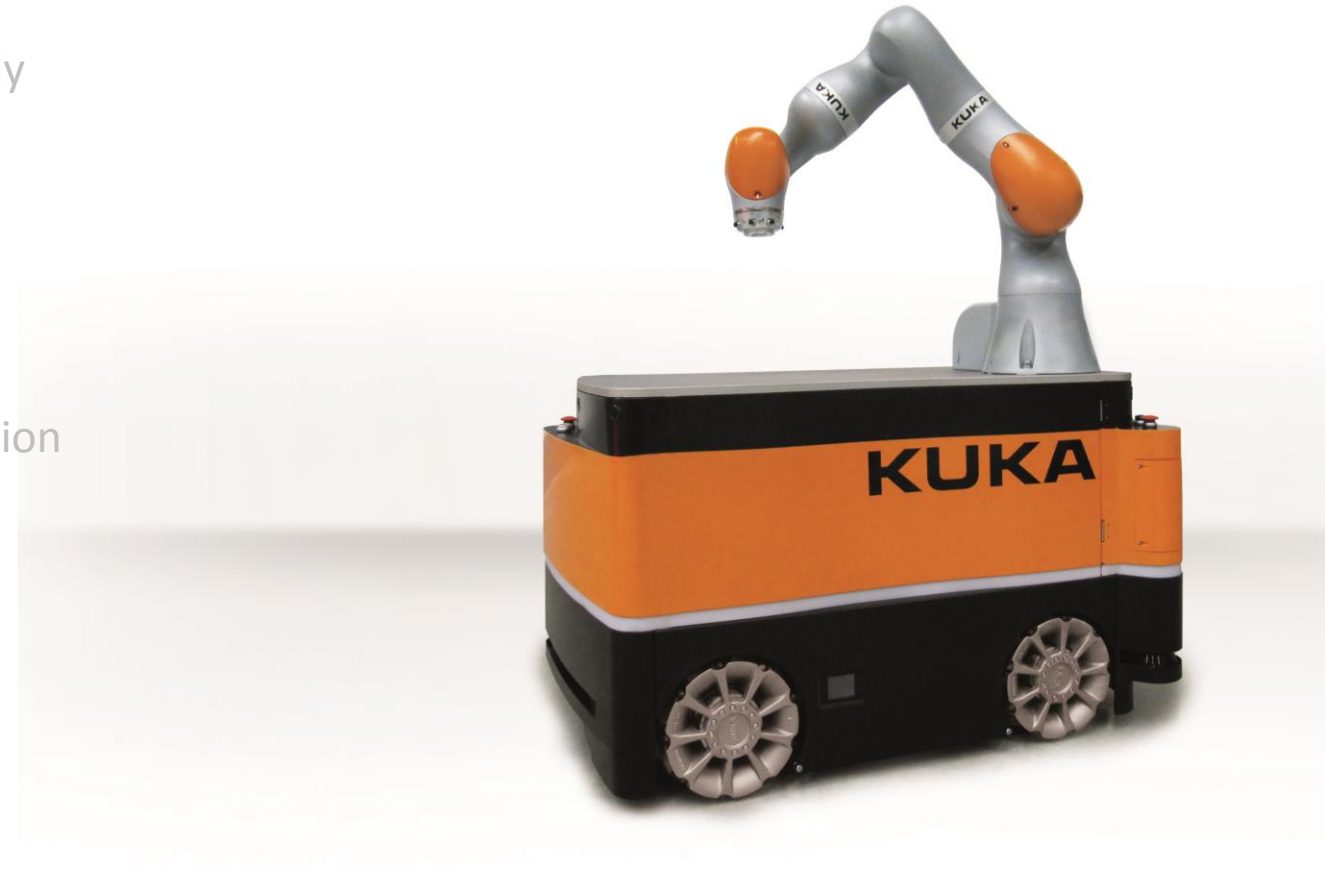


Advantages KMR iiwa in full automation

- Avoid waiting time between process steps
- Increase system capabilities, multi-tasking throw special tooling concepts
- Extend production with new skills / tasks without rebuilding the production line
- Adapt production to needs, products, reduce time to market
- New versatile production concepts
- The KMR iiwa enables costs to be saved in many areas

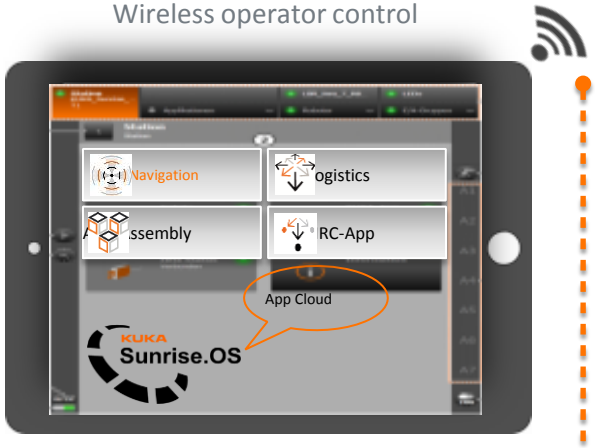
Content

1. Product presentation
   - The new product family
   - Functions
   - KMP omniMove
   - LBR iiwa – Robot arm
   - KUKA Sunrise
   - KUKA Navigation Solution
2. KMR iiwa advantages
3. Industry 4.0



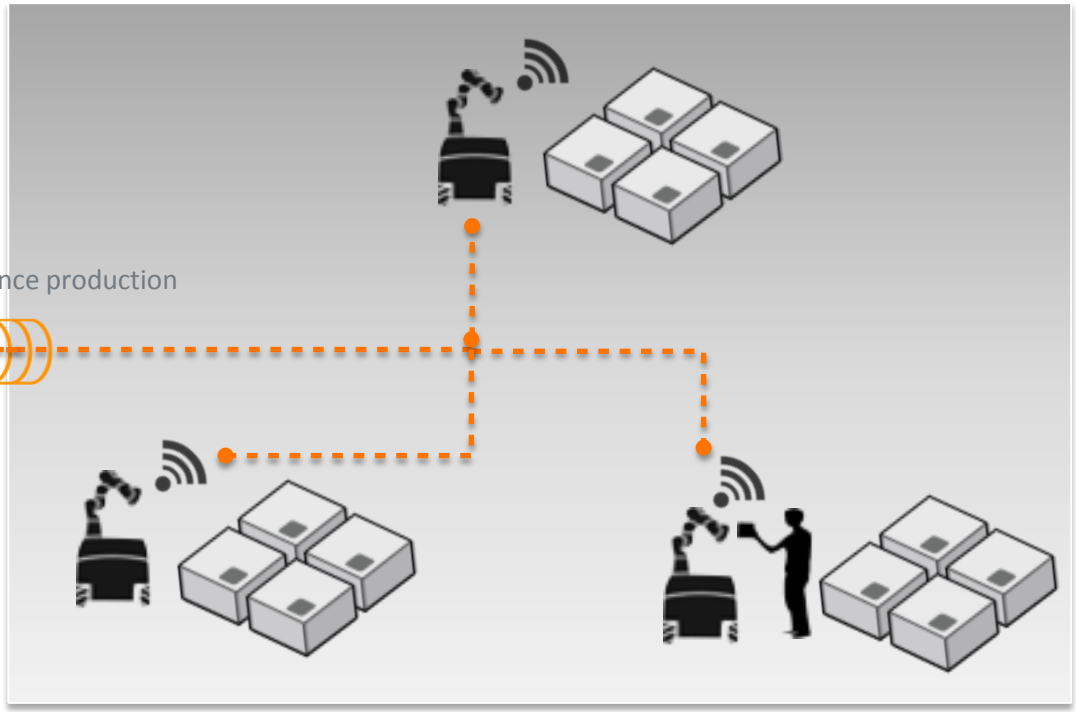
Towards the 4th robotic revolution

Wireless operator control



- Cyber-physical connected systems (simple WIFI connection, Fleet management)

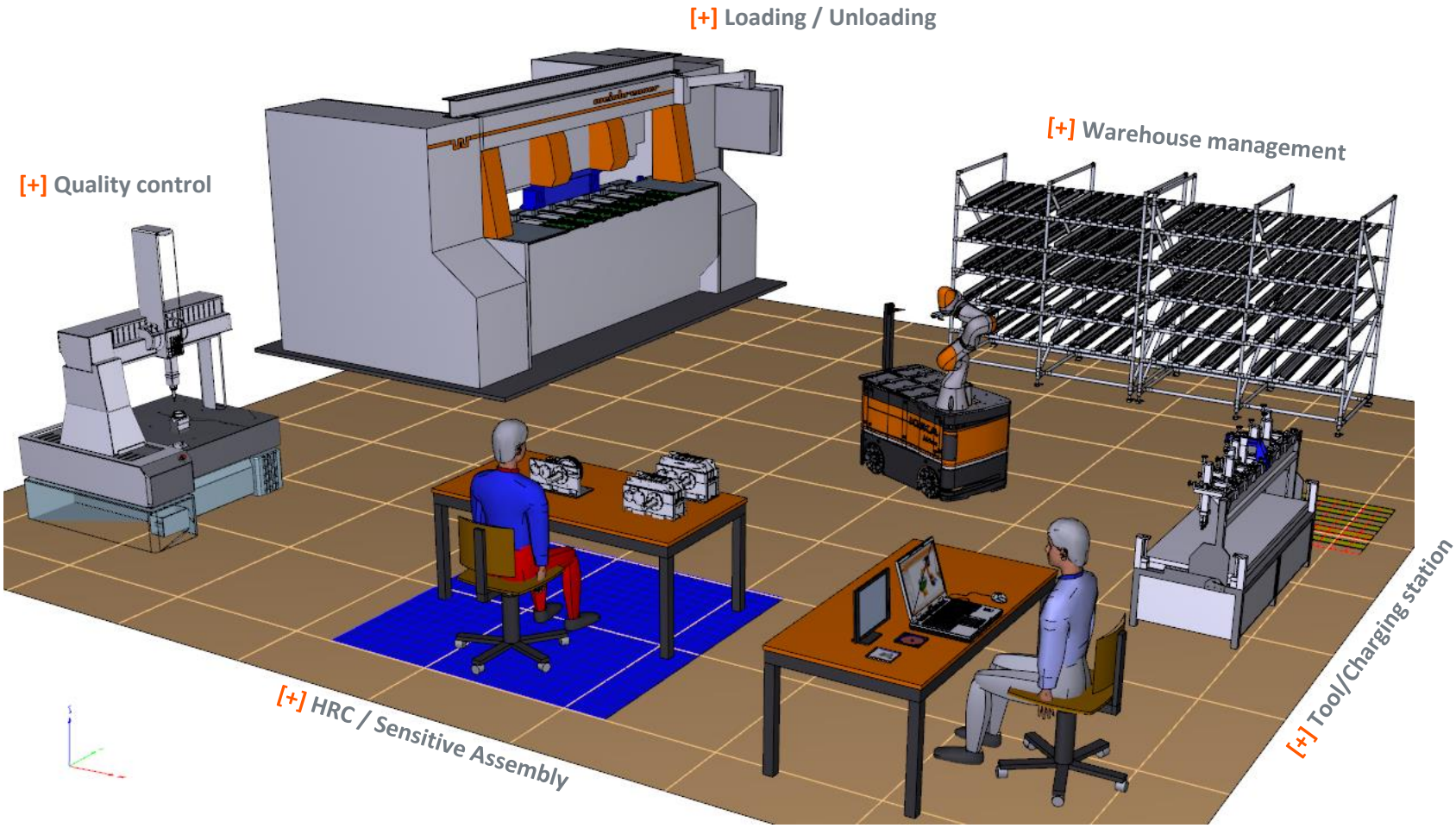
Enhance production



Simple engineering Setup

- Sunrise using mainstream technology (Java, Wireless control…)
- Flexible production concepts, shorten time-to-market

KMR iiwa example application _ multi-tasks _



Mobility and Safety for Industry 4.0

HRC able - Safe work between human and robot

360°

Maximum flexibility and unrestricted maneuverability

Autonomous navigation

Freely scalable modular system

±1 mm

Utmost precision

All driven by one controller
Simple engineering setup

Sensors and auxiliary operator control

Autonomous, sensitive, intelligent for unlimited productivity

Conclusion

”Hello Industrie 4.0 – We go digital

KUKA offers smart solutions for smart factories.
Mobile, safe, flexible”